

MOBILE NETWORK USAGE IN VIETNAM, THAILAND, THE PHILIPPINES, AND MALAYSIA

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Mobile Network Usage in Vietnam, Thailand, the Philippines, and Malaysia

ABSTRACT

This paper proposes a theoretical framework for an enhanced understanding of telecommunications' dynamic growth in Southeast Asia (ASEAN), with a particular focus on the rapid rise in four target markets: Vietnam, Thailand, the Philippines, and Malaysia. Since the introduction of 4G in 2009, 5G subscriptions have been growing, with one-quarter of all mobile subscriptions projected to be 5G by the end of 2024. Globally, 5G is expected to overtake 4G as the dominant mobile access technology by subscription for the first time in 2027. Aiming to assess resources and identify challenges, this study will assist ASEAN countries to leverage mobile networks as a key driver of economic development and social progress, as well as propose a comprehensive exploration of user experience.

This research will initially address the market size and expanding nature of each target market through secondary research methods using country report data. This will form the basis for investigating infrastructure, user behavior and market competition. The next sections will describe these proposed methods and the relevant data collection tools involved. The paper will conclude with a summary of trends and challenges, the key industrial and academic areas that this research could influence.

Key words: Mobile network usage, Telecommunications, Mobile technologies (2G, 3G, 4G, 5G), Southeast Asia (ASEAN)

RESEARCH OBJECTIVES

In order to trace the answer on the performance of mobile network usage in these four ASEAN countries, many interesting avenues that can be explored, particularly Market size, Network resources, User behavior, and Market competition. Research objectives worth exploring include the following:

Research Objective	Research Question	Data Source	Technique & Tool
To study the market size, including subscriber base and revenue, and assess mobile penetration rates; first in Southeast Asia, then focusing on four target markets: Vietnam, Thailand, the Philippines, and Malaysia.	What are the market size and mobile penetration rates in the target markets?	- Country Report - Statista - CEIC	- Descriptive Analysis - Data Collect & Clean & Transform: Excel, PowerQuery - Visualize: Power BI
To study the adoption and deployment of mobile technologies (2G, 3G, 4G, 5G) and assess the network infrastructure in the four target markets: Vietnam, Thailand, the Philippines, and Malaysia.	How are mobile technologies adopted, and what is the state of network infrastructure?	- Country Report - 3G / 4G / 5G bitrates map - Speedtest	- Descriptive Analysis - Data Collect & Clean & Transform: Excel, PowerQuery - Visualize: Power BI
To study the key players in terms of telecommunications in the four target markets: Vietnam, Thailand, the Philippines, and Malaysia.	- Who are the key telecommunications players? - What are their strategies?	- Country Report - Domestic News Article	- Descriptive Analysis - Data Collect & Clean & Transform: Excel, PowerQuery - Visualize: Power BI

Figure 1. Research Questions and Objectives

KEY FINDINGS

1. Market Overview

Experiencing a fast-paced growth, the ASEAN mobile market is characterized by layers of penetration, revenue generation, and growth potential. Zooming in the target markets - Thailand, Vietnam, the Philippines, and Malaysia - these countries appear substantial with a sum of 471 million subscriptions and reach \$22.84 billion in mobile services revenue.

1.1 Population and Mobile Subscription: A Landscape of Disparities and Saturation

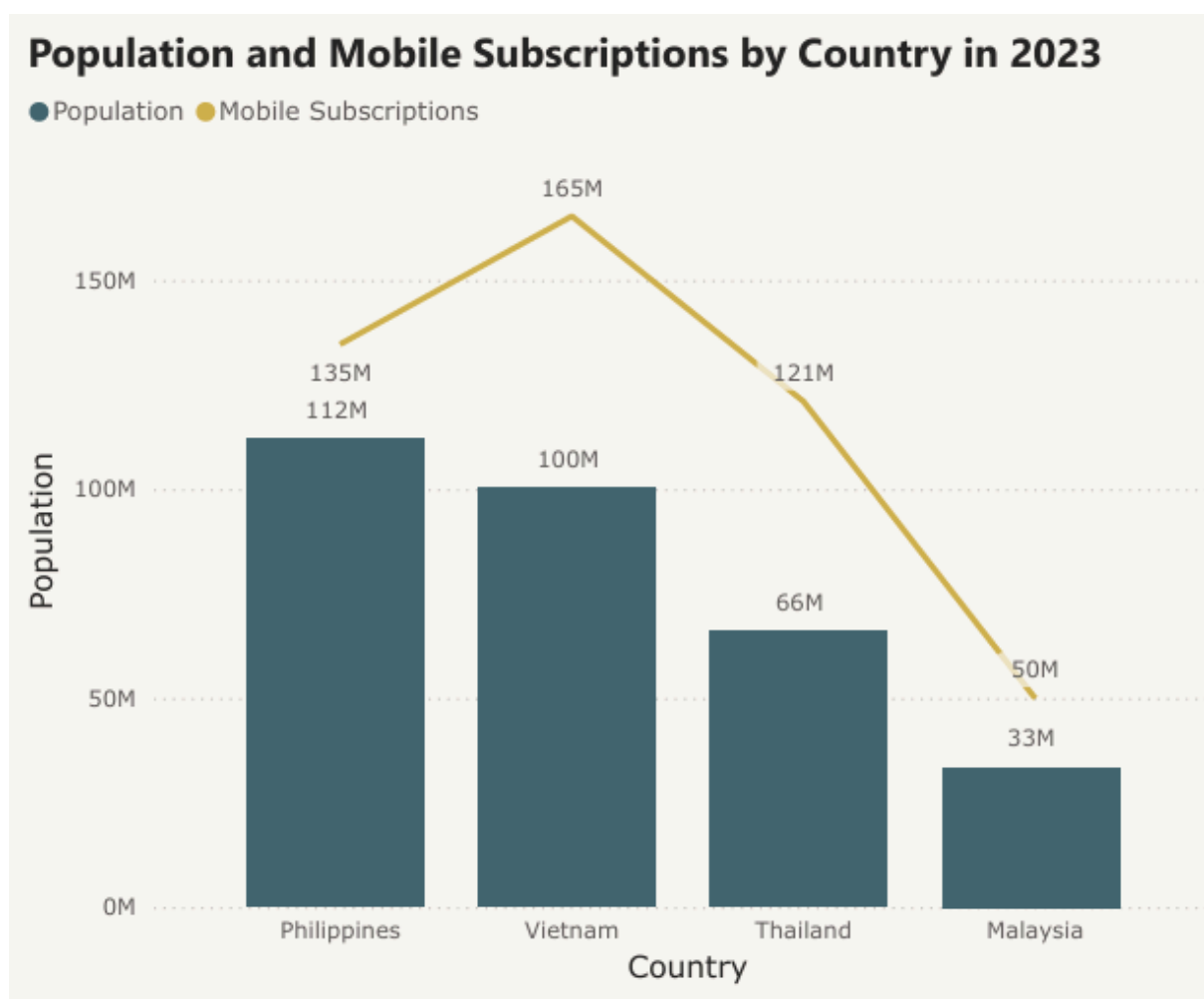


Figure 2. Population and Mobile Subscriptions by Country in 2023

Figure x illustrates the population, mobile subscriptions, and penetration rates of the four countries. Firstly, with a population of 100 million (GSO 2023), Vietnam holds a peak position in mobile subscriptions with 165 million (MIC 2023). Therefore, the gap between these two numbers suggests that many people in Vietnam use more than one SIM card. The Philippines experiences a slower trend, with a smaller gap between the subscription rate and population, having 135 million subscribers (Statista 2023b) and 112 million people (Statista 2023a), respectively. Additionally, Thailand shows a high ratio with 121 million subscriptions (CEIC n.d.) and a population of around 66 million (BOI n.d.). Finally, Malaysia has around 50 million subscriptions (Statista 2023c) with a population of 33 million (DOSM n.d.), which is a moderate ratio.

Country	Mobile Penetration Rate (Subscriptions / Population, %)
Thailand	183%
Vietnam	165%
Malaysia	150%
Philippines	120%

Figure 3. Mobile Penetration Rate by Country in 2023

The Penetration Rate indicates the ratio of customer usage compared to the potential customer base. When the Penetration Rate is higher than 100%, it means that there are more active mobile subscriptions than the total population. According to Figure X, all countries' rates exceed 100%, with Thailand leading at a remarkable 183% penetration rate, followed by Vietnam (165%), Malaysia (150%), and the Philippines (120%).

According to two figures, four markets are experiencing a saturated trend where individuals commonly use multiple SIM cards. This presents opportunities for telecommunications brands to market through promotions, expand network coverage areas, and offer tailored data plans for specific needs. On the other hand, an industry with a high penetration rate also means that the potential customer base is now fully equipped with the service. As a result, telecommunications companies might have to shift their strategy from acquiring new customers to enhancing customer loyalty.

1.2 Mobile Services Revenue and CAGR: Growth Potential and Market Maturity

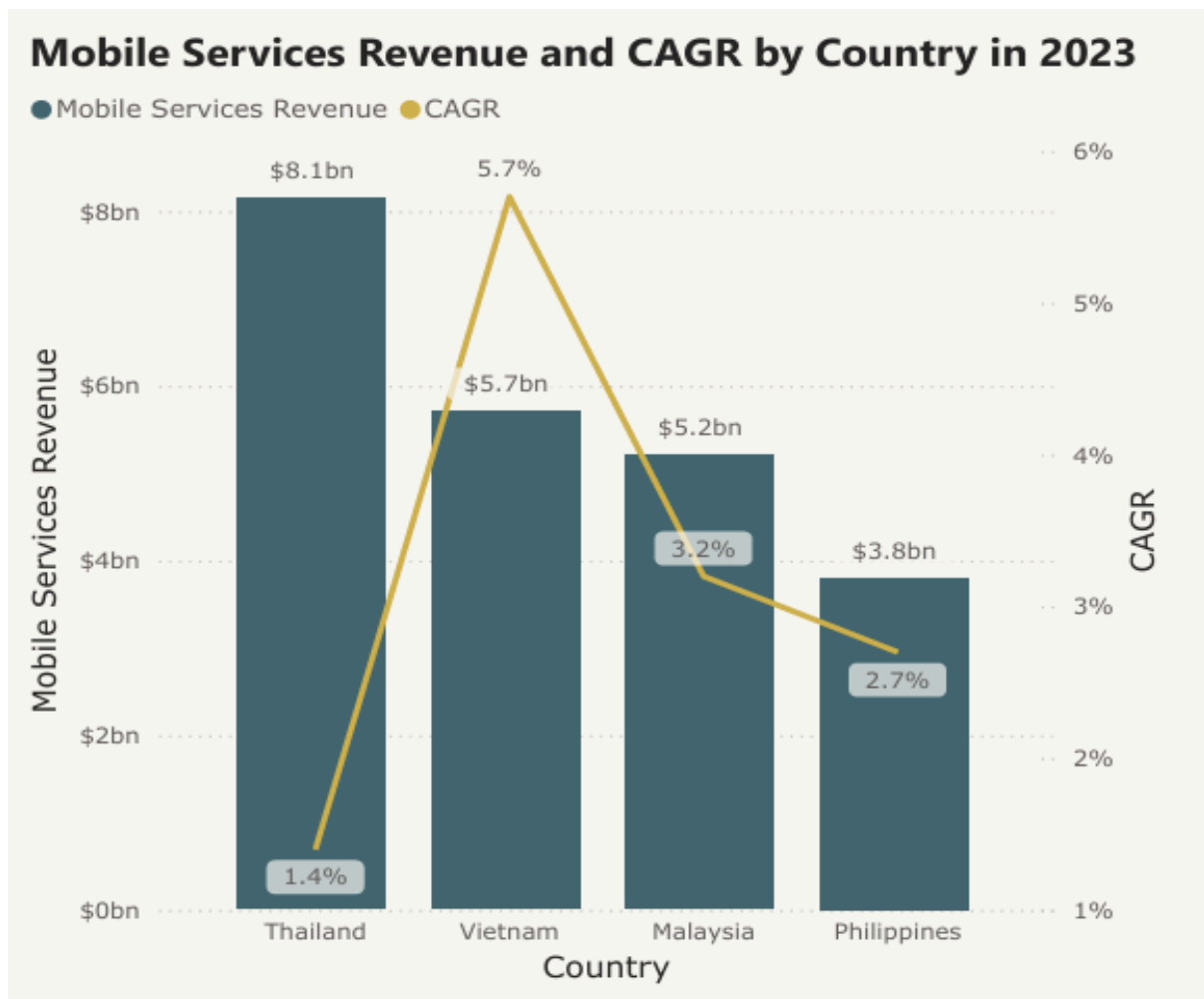


Figure 4. Mobile Services Revenue and Compound Annual Growth Rate (CAGR) by Country in 2023

To better understand the financial performance and future growth prospects in each of the four markets, this section will examine the relationship between revenue and Compound Annual Growth Rate (CAGR). With a combination of a high penetration rate and potentially higher average revenue per user (ARPU), Thailand leads in mobile services revenue despite its relatively small population. However, a low CAGR of 1.4% indicates a mature market with limited room for organic growth.

Besides, having a smaller revenue compared to Thailand - Vietnam, at \$5.7 billion, emerges the peak in CAGR (5.7%). This number states a strong growth potential, backed up by a large population, increase in digital usage and potential for service expansion when the middle class rises.

The last two countries have moderate revenue figures (\$3.8 billion and \$5.2 billion, respectively) and medium CAGRs (2.7% and 3.2%), suggesting that these markets still

have growth potential. Initially, these two markets need a strategic approach in expanding network infrastructure, introducing innovative services, and offering tailored solutions.

2. Network Infrastructure

2.1. Technology Adoption and Coverage

In this industry, technology adoption among types of mobile networks (2G, 3G, 4G, 5G) plays an influential role in affecting network usage. Therefore, by analyzing the spread of technology in each country, this study will touch upon valuable insights into the growth of these mature mobile landscapes.

Country	2G	3G	4G	5G
Vietnam	Phased out by Oct 2024	Expected phase-out by 2028	99.4% population coverage	Some large city, expected to cover 99% of the population by 2030
Thailand	Phased out by Dec 2021	Available	99.5% population coverage	95% population coverage
Philippines	Available	Available	80% population coverage	6% population coverage
Malaysia	Available in some rural area where 3G and 4G coverage is limited	Phased out by 2021	97% population coverage	82.1% population coverage ⁴²

Figure 5. Summary of Technology Adoption in 2024

In Vietnam, the fast-paced adoption of newer mobile technologies is a notable trend. The country deactivated 2G in October 2024 (Vietnamnet 2023) and plans to phase out 3G by 2028 (SGGP News 2024). Notably, 99.4% of the population now has access to 4G (Vietnamplus 2024), which serves as a backup to the decisive shift and is becoming the dominant network. According to VNEconomy (2024), 5G has been available in some key cities since 2023, with a plan to achieve 99% coverage by 2030 throughout the country. This master plan indicates strong growth and the potential for a tech-forward approach to digital infrastructure development.

Thailand - the leader in technology adoption - successfully deactivated 2G in December 2021 (TNSi 2023), marking a milestone in developing a more sophisticated network in terms of experience. Nationally, 3G is still available, while with near-universal coverage of 99.5% as of Q2 2024 (GSMA 2024), 4G is the dominant network. Impressively, 5G now reaches 95% user coverage in Thailand (The Fast Mode 2024), positioning the country at the forefront of network infrastructure development.

Data from the Philippines shows an opposite trend of slow transition across mobile networks. According to StaffDomain (2024), this country still maintains the availability of both 2G and 3G, since a huge segment of rural citizens have a strong reliance on these technologies due to the weak 4G signals in non-urban areas and time device replacement cycles. Noticeably, 4G is the dominant technology - 80% coverage among Filipinos (Statista 2024, Newsbytes 2024). 5G coverage, on the other hand, is significantly low - 6%, which initially highlights the early stage of adoption. The Philippines is still in an early development phase, with much room for future growth.

Malaysia demonstrates a steady development towards newer mobile technologies. In fact, 2G is still a key service in rural areas, the country has successfully launched 3G phase out in December 2021 (Kore Wireless 2024). According to Sony (n.d.), the key reasons behind 2G persisting while 3G is being sunsetted is due to the 2G's cost-effective option for basic needs for rural citizens while many legacy technologies, namely Internet of Things (IoT)-related, have a strong reliance on 2G. At the same time, the country boasts 97% 4G coverage, making it the primary mobile technology in use (Yahoo Finance 2024). Malaysia's 5G coverage in populated areas reached 82.1% as of October 2024 (The Sun Daily 2024). With rapid expansion, Malaysia maintains a proactive approach to updating and deploying the latest generation of mobile networks.

The following coverage maps illustrate the geographical reach of mobile networks in each country.

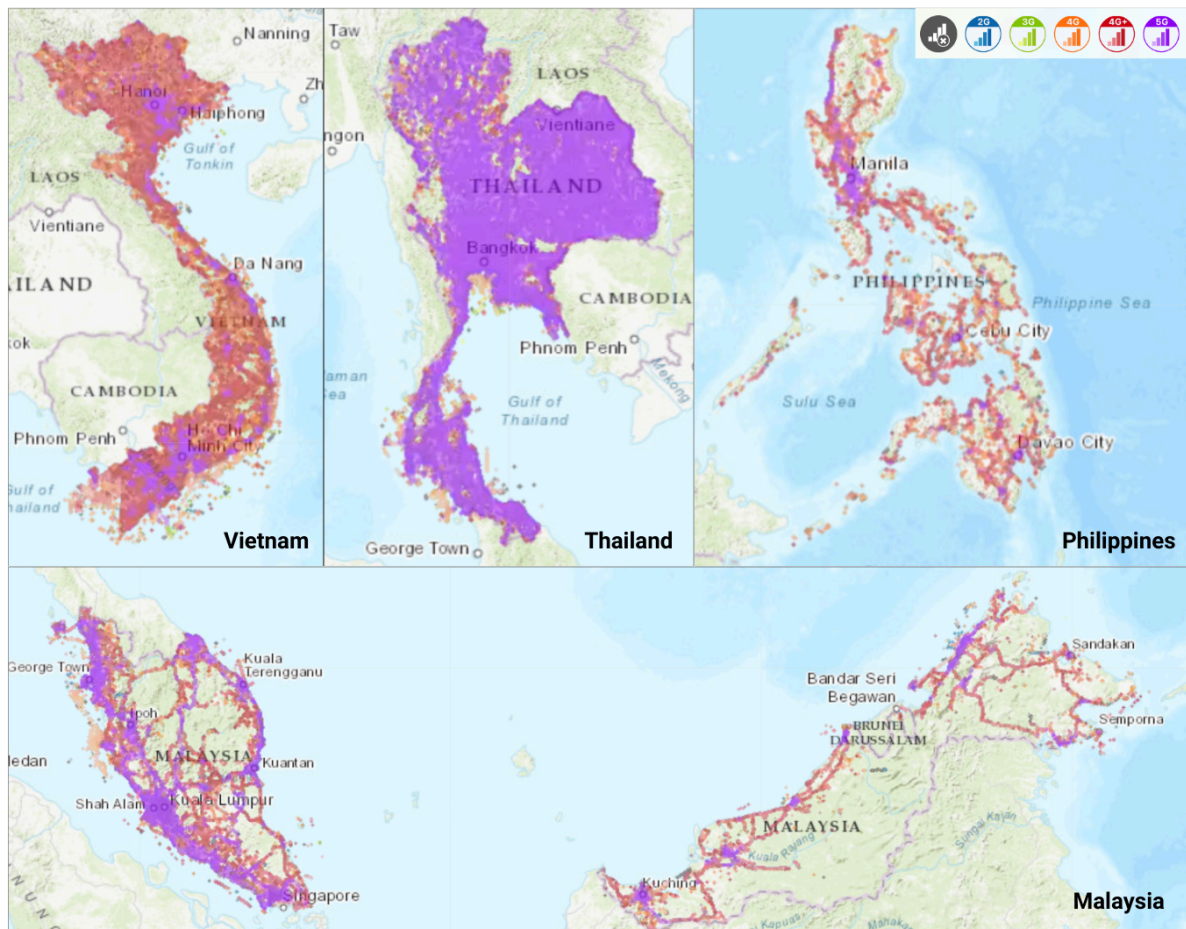


Figure 6. Network Coverage Map in 2024 (Nperf)

This figure shows a diverse mobile landscape across Southeast Asia. In Vietnam, it is evident that higher technologies provide more coverage along the economically vital coastline and extends deep into the fertile Red River and Mekong deltas. These areas can also be considered the nation's economic growth points. Central areas, however, experience a limited penetration rate due to the geographical location and difficult terrain. This indicates the necessity for a strategy to overcome the dual challenges of rugged terrain, lower population density and the deployment of technology in these regions.

Upon examining the data from Thailand, it is evident that there is near-uniform coverage blanketing the entire country. This extensive network indicates a robust infrastructure and significant efforts to bridge the urban-rural gap in terms of economy and technology. This achievement is largely due to the country's flat topography. Considering the complexities of the Philippines' archipelagic geography, with its many small islands, advancing technology presents a significant challenge. Noticeably, the

coverage network is fragmented, clustering around major urban centers on various islands, while many smaller islands and inland areas remain underserved. This highlights a significant rural-urban divide, resulting in a pronounced digital gap.

Finally, two trends can be observed through the data from Malaysia. First, comprehensive coverage dominates Peninsular Malaysia, especially along the west coast corridor, an economically vital area. On the other hand, East Malaysia, with its challenging terrain and dispersed population, experiences a lower coverage ratio. These two distinct growth patterns result from differences in geography, economic priorities, and strategic infrastructure development in each part of the country.

2.2. Mobile Network Quality

In the mobile landscape, besides the foundational aspects of technology adoption and network coverage, network quality is a key dimension for end-users. Therefore, in this section, network quality will be examined through key performance indicators (KPIs), including median download speed, median upload speed, download/upload speed ratio, and global speed ranking.

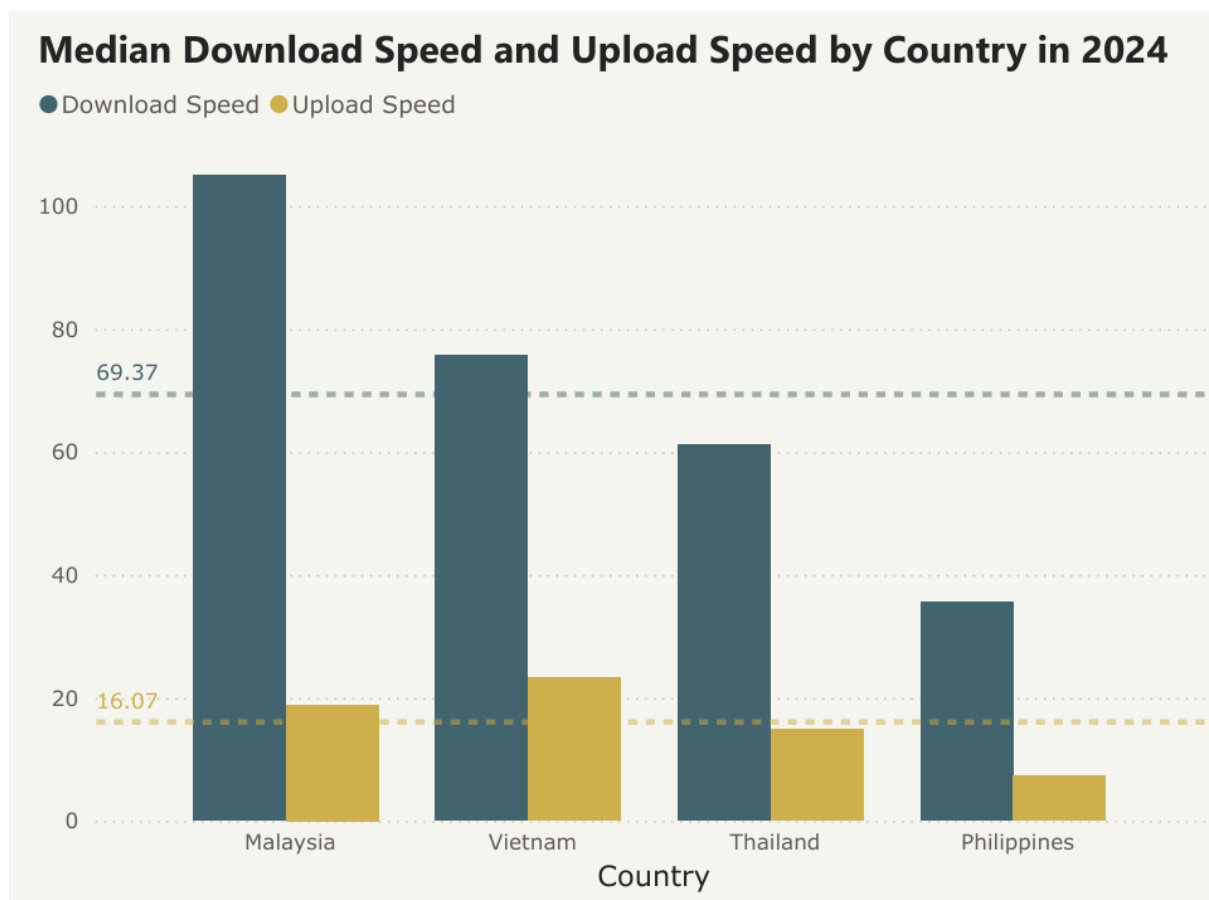


Figure 7. Median Download Speed and Upload Speed by Country in 2024

Country	Download/Upload Ratio	World Speed Ranking
Philippines	4.82	76
Thailand	4.12	49
Vietnam	3.26	44
Malaysia	5.57	20

Figure 8 . Download/Upload Ratio and World Speed Ranking by Country in 2024
(Speedtest 2024)

With a median download speed of 91.49 Mbps and a 20th position in the global ranking, Malaysia leads due to its advanced network infrastructure. Thailand (51.14 Mbps download, 49th globally) and Vietnam (47.66 Mbps download, 44th globally) highlight similar yet competitive performance with balanced ratios, indicating a high level of user experience. In the Philippines, the data is somewhat behind (35.38 Mbps download, 76th globally) with a download/upload ratio of 4.82, which is linked to the geographical challenges previously discussed. Malaysia's ratio of 5.57 suggests a network optimized for download traffic, while Vietnam's lower ratio of 3.26 indicates a more balanced, though potentially less performant, network.

Differences in network quality directly affect user experience and digital fairness. In Malaysia, higher speeds mean smoother streaming and faster downloads. In contrast, lower speeds in the Philippines can lead to poor app performance. This data highlights the importance of ongoing infrastructure investment, good spectrum management, and a competitive market in achieving high network quality.

3. Competitor Landscape

3.1 Vietnam

Vietnam is a dynamic market dominated by three key players: Viettel, MobiFone, and VinaPhone (MIC 2024), each of which prospect unique strengths, weaknesses, and strategic approaches to attract and retain customers in a rapidly evolving technological environment.

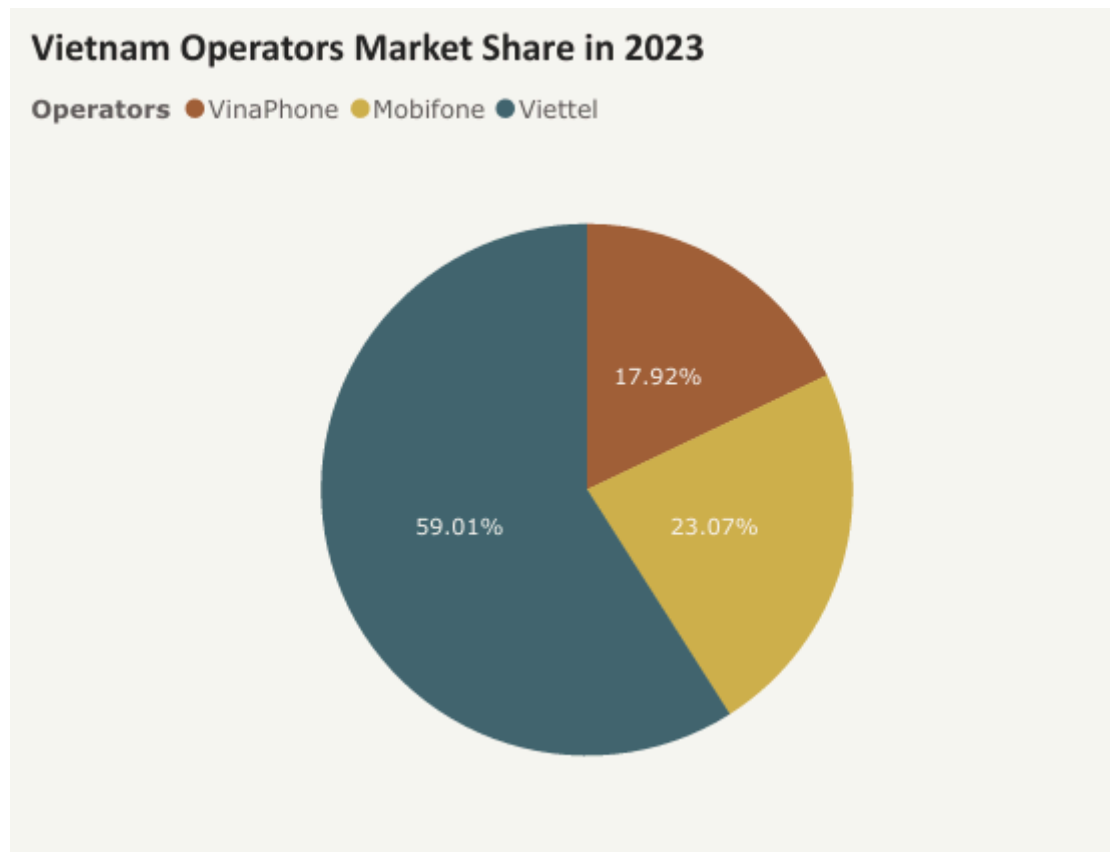


Figure 9 . Vietnam Operators Market Share in 2023

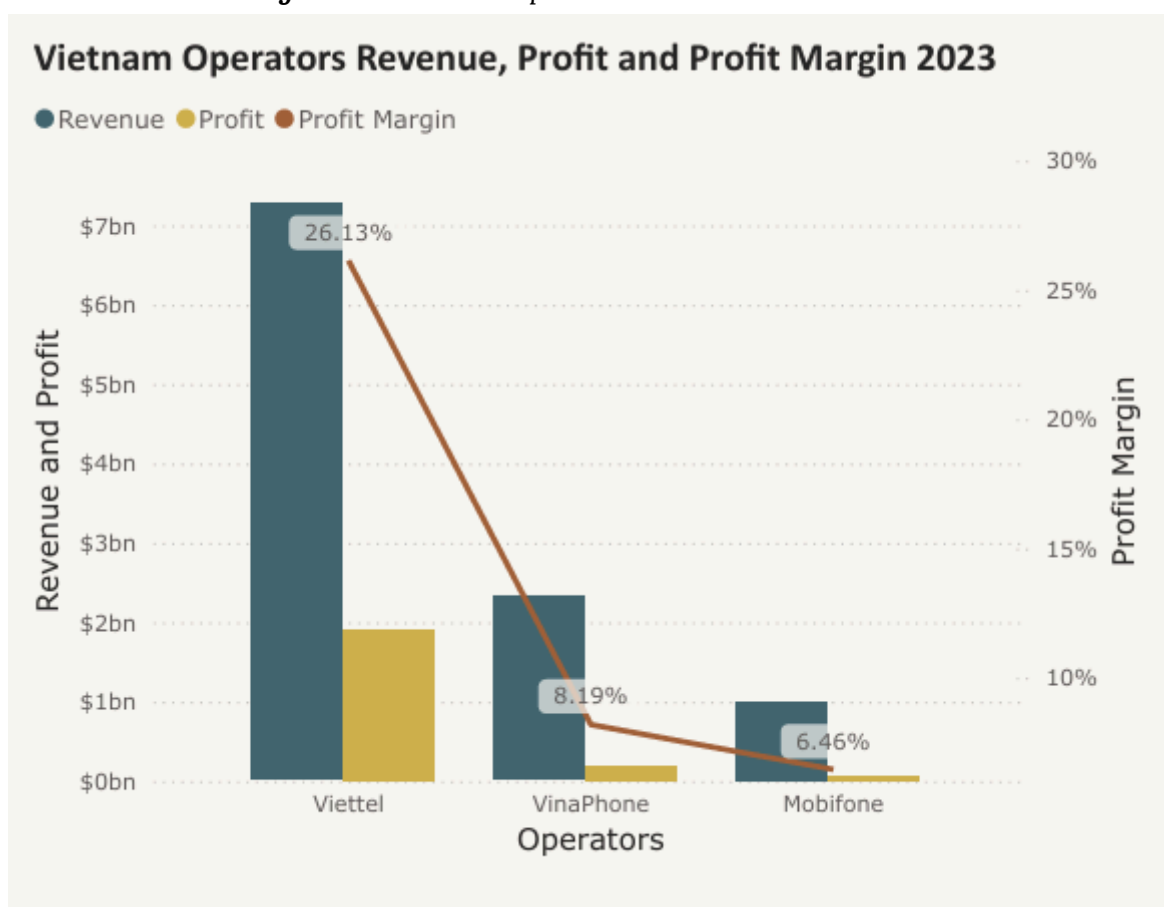


Figure 10. Vietnam Operators Revenue, Profit and Profit Margin in 2023

Viettel – market leader, had over \$7 billion revenue (VNExpress International 2023) with 26.13% of profit margin, is significantly outperforming the market. The strength in financial position assists the company in its network infrastructure investment, technological advancements, and value-added services, which are potential initiatives to consolidate its leading position. In terms of mobile services, Viettel having a focus on premium offerings and its extensive 4G/5G coverage contribute to its ability to command a price premium and achieve higher profitability (Lim 2018).

With substantial revenue in 2023 (\$2.2 billion) (Tuoi Tre 2024), VinaPhone lags behind the previous company with a profit margin of only 8.19%. This figure indicates challenges in cost management and bundled-services offerings. Despite having the VNPT technology ecosystem as its backbone (VNPT 2024), VinaPhone still needs to strengthen its differentiation strategy and improve operational efficiency to achieve sustainable growth in this market.

MobiFone, with a revenue of \$1 billion and a profit margin of 6.46%, is facing the toughest profitability challenges among the three operators. Its focus on keeping prices low and targeting urban markets might be limiting its potential for revenue growth. Additionally, its slower adoption of 5G technology could make it harder to attract high-value customers. To improve its financial performance, MobiFone needs to explore new ways to generate revenue and find ways to cut costs.

These financial indicators show how competitive the Vietnamese mobile network market is. With a significant potential in financial position, Viettel has the ability to invest heavily in expanding its network and innovating its services, putting pressure on VinaPhone and MobiFone to improve their offerings and operational efficiency. This competition is likely to lead to more innovation, which could benefit consumers with better service quality, lower prices, and more value-added services.

3.2 Thailand

After the merger of Dtac with TrueMove H in early 2023, the Thai market has become intensely competitive, shaped by two key players: AIS and TrueMove H. These two companies differentiate themselves through diverse strategies to cater to the evolving needs of Thai consumers.

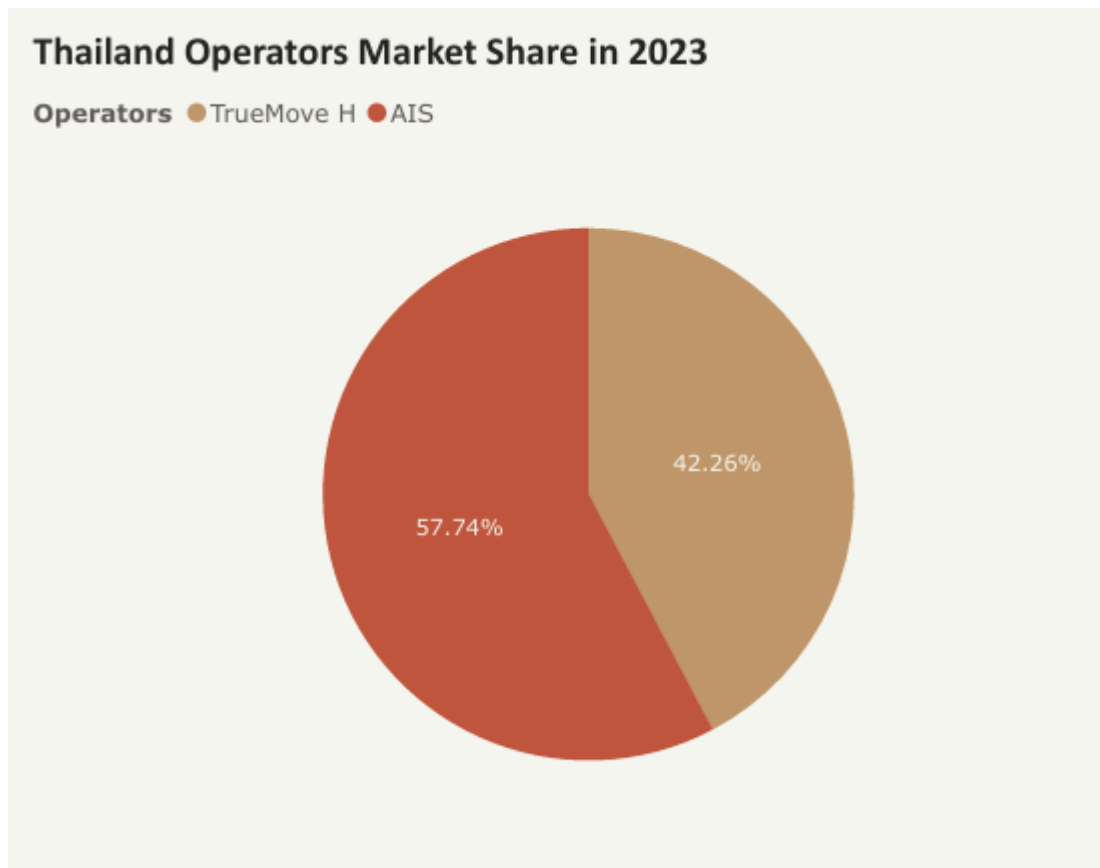


Figure 11. Thailand Operators Market Share in 2023

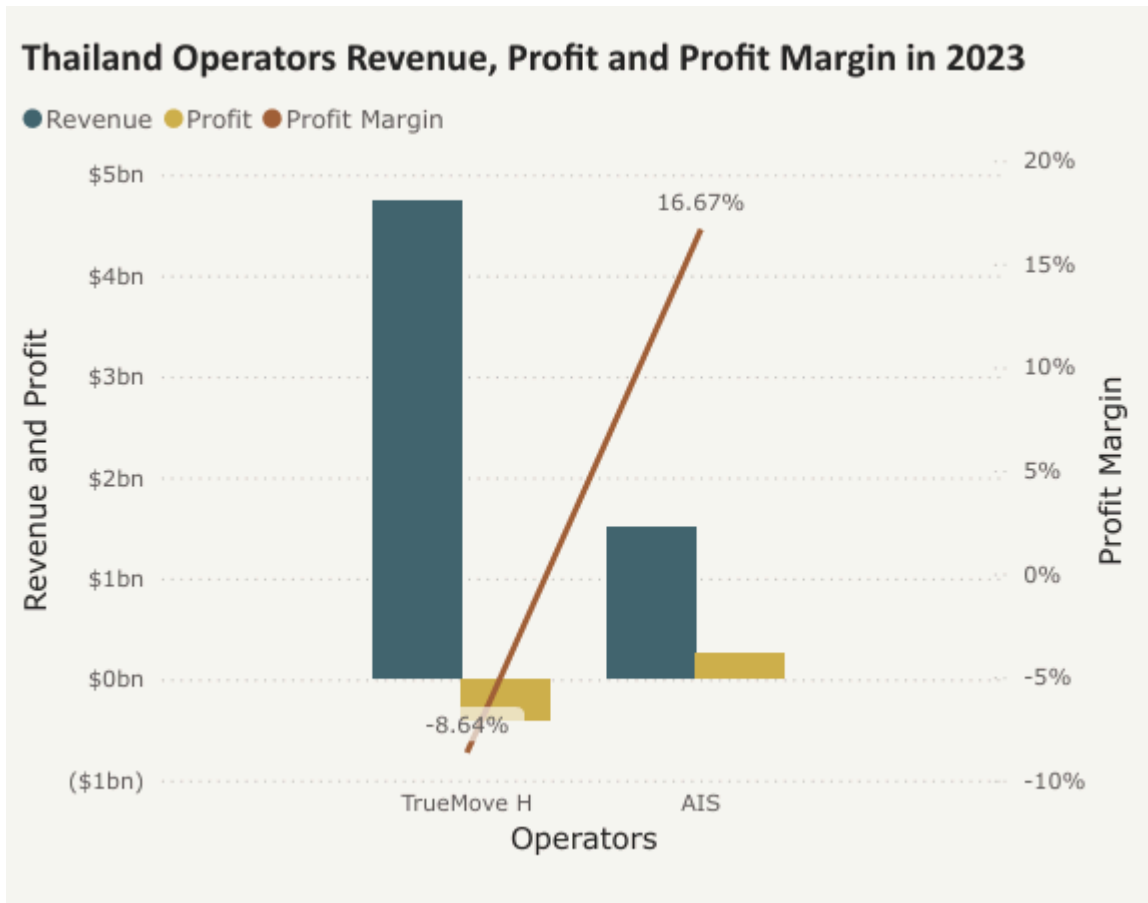


Figure 12. Thailand Operators Revenue, Profit and Profit Margin in 2023

AIS, the largest mobile operator in Thailand, has extensive network coverage and is known for high-speed internet, even in challenging areas like mountains and islands. This strong network is supported by innovative services and digital content, including partnerships with streaming platforms and entertainment providers. AIS positions itself as a premium provider, offering a range of prepaid and postpaid plans to a broad customer base. According to the Bangkok Post (2024), AIS had the highest revenue among the operators in 2023, with a healthy profit margin of 16.67%. This shows that AIS has successfully balanced premium pricing with attractive services to maintain a strong customer base and profitability.

TrueMove H, following its merger with Dtac in early of 2023, has solidified its position as a major player in the Thai market. This strategic move has expanded its subscriber base and network capacity, allowing it to compete more effectively with AIS. TrueMove H, a subsidiary of True Corporation, offers a comprehensive suite of services, including mobile, broadband, and digital content. It leverages its parent company's ecosystem to provide bundled packages and integrated services, such as TrueID for entertainment and TrueMoney for financial services. The company focuses on delivering value and convenience to customers, with competitive pricing and a strong emphasis on digital

lifestyle offerings. However, Figure 2 reveals that despite generating considerable revenue, TrueMove H is struggling with profitability, reporting a negative profit margin of -8.64% (Market Screener 2024). The integration of dtac's operations into TrueMove H has likely incurred significant costs, including network integration, rebranding, and streamlining operations. These one-time expenses can put a strain on profitability in the short term.

The Thai mobile network market is highly competitive, influenced by factors such as network quality, pricing, data allowances, and value-added services. Operators are continuously innovating to attract and retain customers, focusing on expanding 5G coverage, enhancing digital content offerings, and developing personalized services. The rising demand for data-driven services, fueled by the growing popularity of streaming platforms and online gaming, is further intensifying the competition between AIS and TrueMove H.

3.3 The Philippines

The Philippine mobile network market is dominated by three major players: Globe Telecom, Smart Communications, and Dito Telecommunity. Each company caters to various customer needs with distinct strategies, strengths, and weaknesses, creating a vibrant and competitive environment.

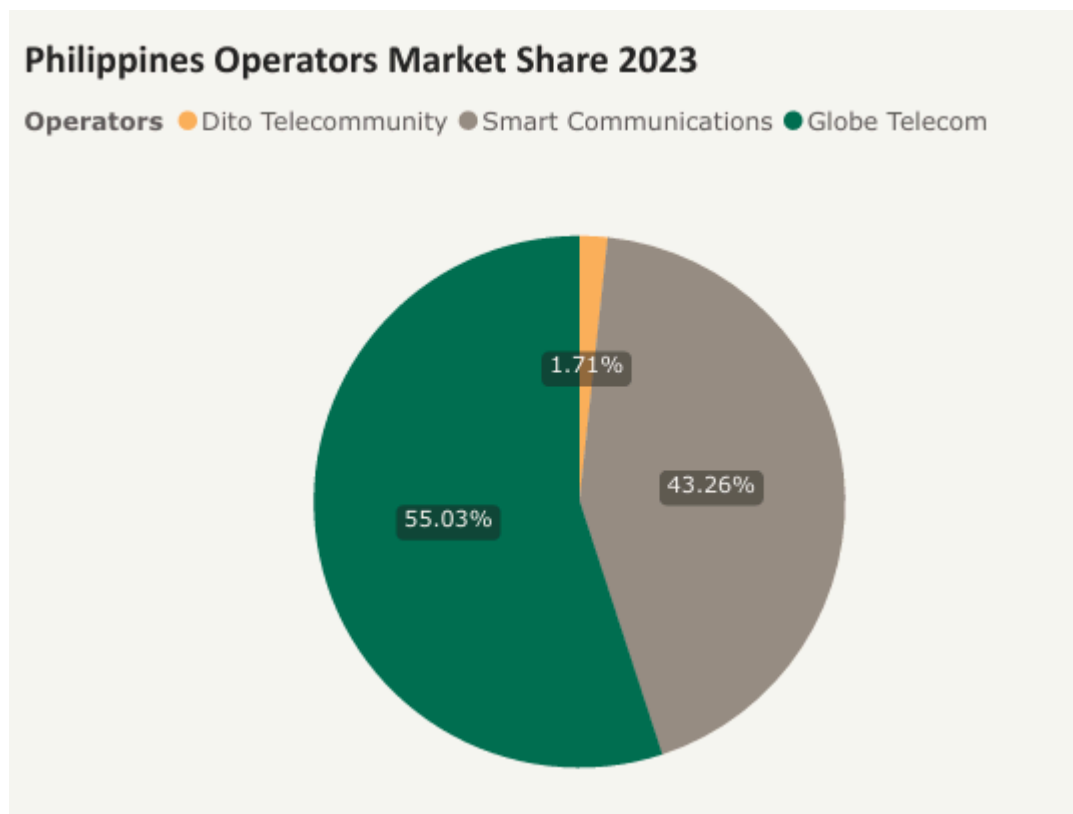


Figure 13. Philippines Operators Market Share in 2023

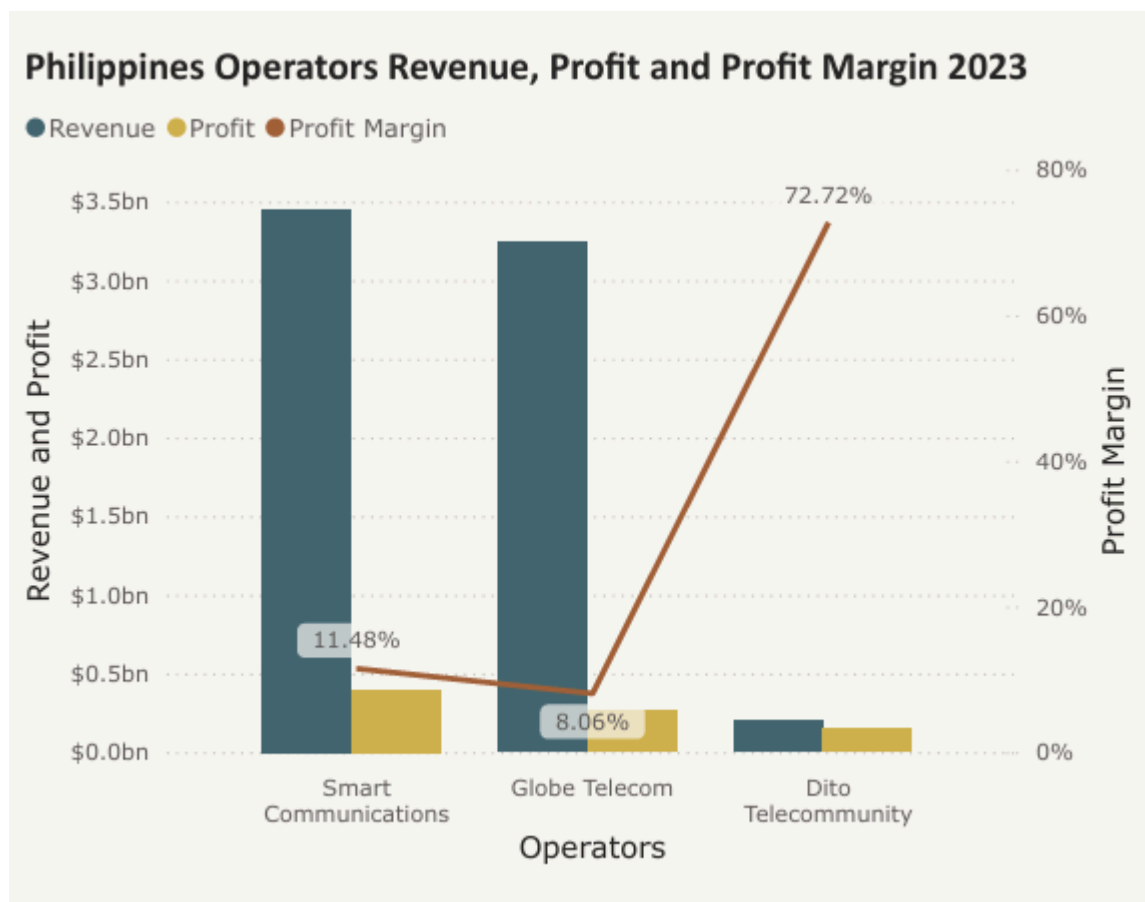


Figure 14. Philippines Operators Revenue, Profit and Profit Margin in 2023

Globe Telecom, part of Ayala Corporation and backed by Singtel, is a top provider known for its quality and extensive 4G LTE and 5G coverage. Globe offers bundles that include mobile, broadband, and digital financial services through GCash, as well as entertainment options like Disney+ and Netflix (Globe Telecom 2023). This approach helps Globe attract a wide range of customers, from prepaid users to high-end postpaid customers, mainly in urban areas. Globe's focus on quality and lifestyle integration, along with strong partnerships with content providers, makes it a premium service provider. However, Globe faces challenges with network congestion and higher prices compared to competitors. Despite generating \$3 billion in revenue, its profit margin of 8.06% is lower than Dito's (InsiderPH 2024), indicating higher operating costs that affect overall profitability.

Smart Communications, a subsidiary of PLDT Inc., boasts a nationwide fiber and wireless network, offering high mobile internet speeds and a strong presence in enterprise connectivity and fintech services (via PayMaya). Smart targets a wide audience, from budget-conscious prepaid users to high-end postpaid customers, offering competitive pricing and value-added services like entertainment perks and

flexible data plans. Smart's strength lies in its established infrastructure and diverse offerings, but it faces challenges with customer service and service interruptions, often perceived as reactive to Globe's initiatives. Figure 2 illustrates Smart's strong revenue generation (\$3.5 billion) (First Pacific 2024), coupled with a reasonable profit margin of 11.48%. This indicates a balanced approach to pricing and service offerings, allowing Smart to maintain a competitive edge while generating profits.

Dito Telecommunity, a newer player in the market, is a joint venture between Udenna Corporation and China Telecommunications Corporation. Dito stands out with its aggressive pricing and focus on affordability, targeting cost-conscious users with simple and transparent bundled services. Its competitive pricing and emphasis on new technologies attract younger audiences and digital natives. However, Dito struggles with limited network coverage, especially in rural areas, and relies on rapid expansion for growth. Interestingly, despite being the newest player, Dito has the highest profit margin (72.72%) among the three operators, even though it has the lowest revenue (\$0.5 billion) (Inquirer Business 2024). This indicates that Dito's aggressive pricing strategy and efficient cost management enable it to achieve high profitability despite a smaller revenue base.

Competition in the Philippine mobile network market is fueled by network quality, pricing, data allowances, and additional services. The rising demand for data services like streaming, gaming, and digital finance is intensifying this competition. To attract and retain customers, operators are expanding 5G coverage, enhancing digital content, and offering personalized services.

3.4 Malaysia

The Malaysian mobile network market is very competitive, with three main companies: Telekom Malaysia (TM), Celcom Digi, and Maxis. Each has its own strengths and weaknesses and uses different strategies to attract and keep customers in a fast-changing tech environment.

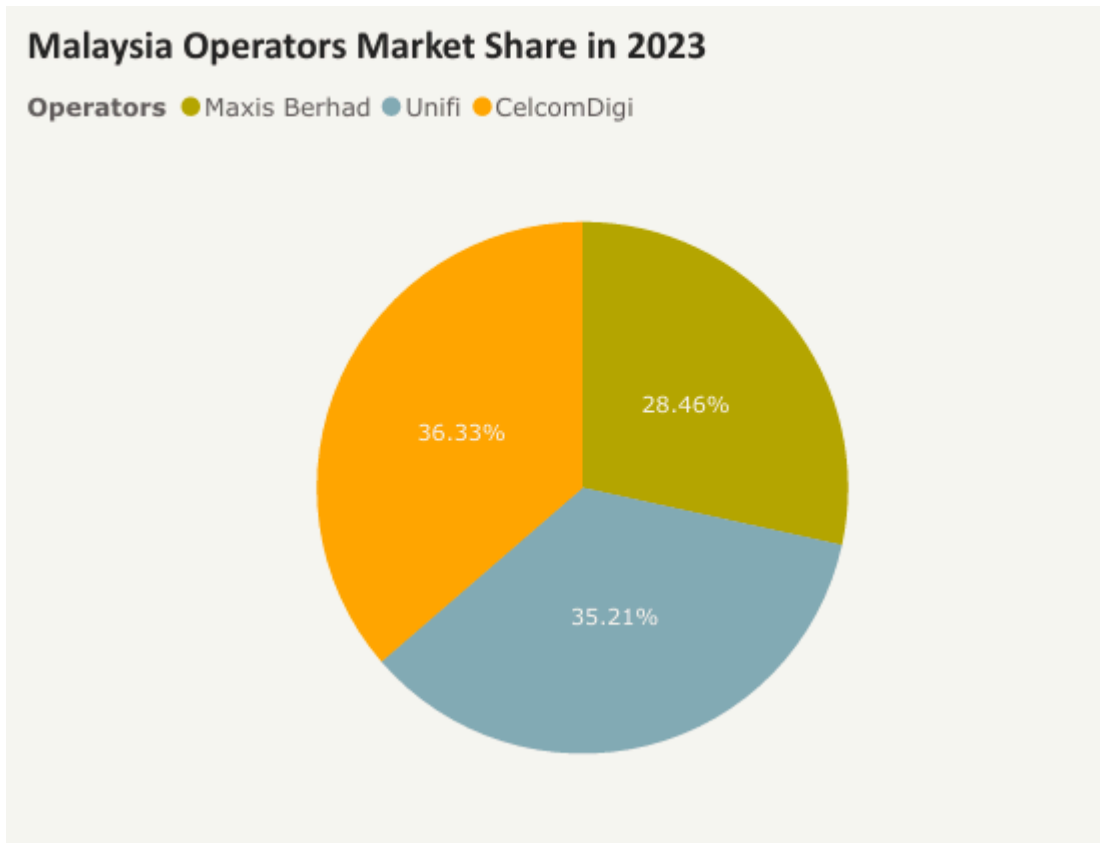


Figure 15. Malaysia Operators Market Share in 2023

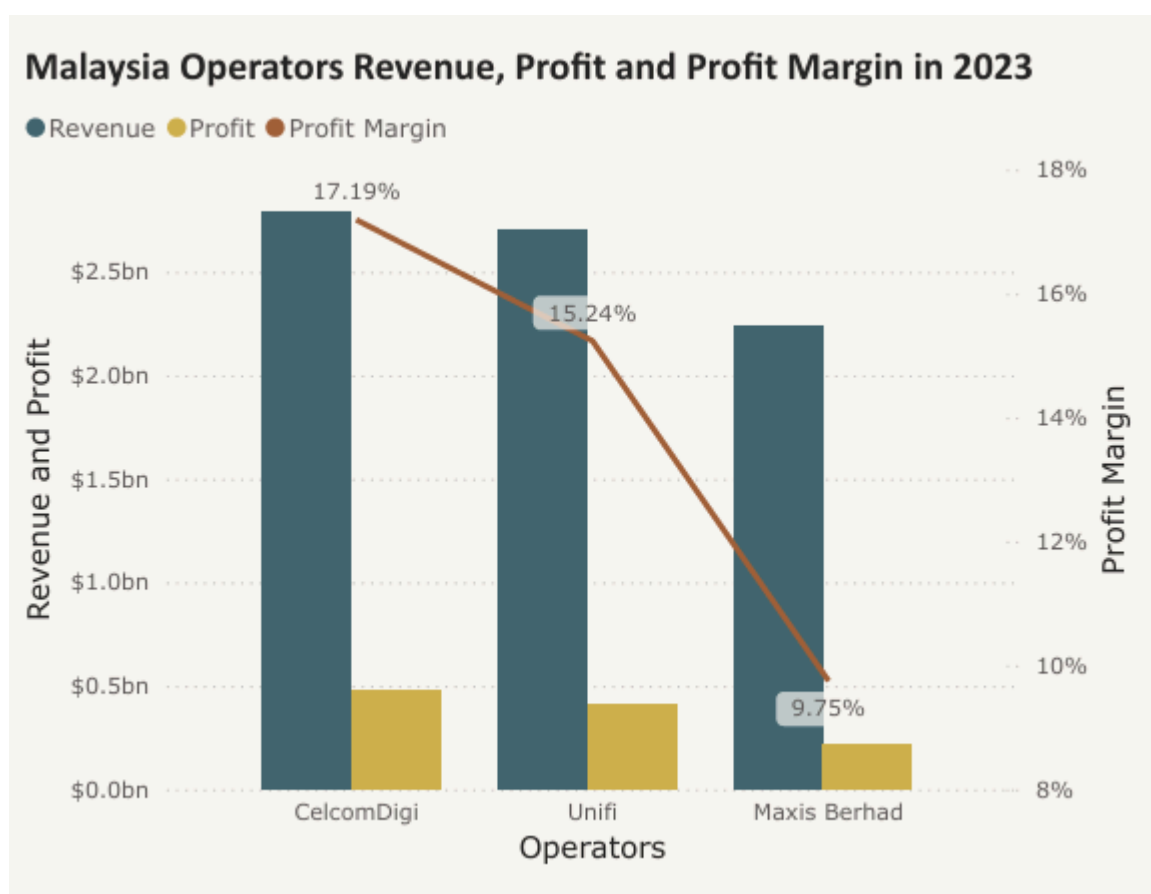


Figure 16. Malaysia Operators Revenue, Profit and Profit Margin in 2023

Digital, broadband, and fixed-line services are the areas of expertise for TM, a prominent integrated telecommunications provider. TM maintains a competitive edge, especially in fixed-line broadband and digital services for home and business users, by utilizing its solid fixed-line infrastructure and government support. But compared to its rivals, TM is less well-known in the mobile space, which would reduce its overall market share. This is reflected in Figure 2, which shows that although TM (Unifi) makes a good \$2.7 billion in revenue (TM 2023), its profit margin of 15.24% is less than Celcom Digi's, suggesting that it may be difficult to compete successfully in the mobile market.

In 2022, Celcom and Digi merged to establish Celcom Digi, which now has more subscribers than any other mobile network operator in Malaysia. A powerful player with competitive pricing and wide network coverage has been produced by this combination. With a variety of data and mobile plans, Celcom Digi caters to both postpaid and prepaid consumers with alluring promos and value-added services. It may take some time to develop a strong, cohesive corporate identity, and the company has difficulties in completely merging the networks and operations of Digi and Celcom. Notwithstanding these difficulties, Figure 2 shows Celcom Digi's impressive financial results, including the greatest revenue (\$2.8 billion) (CelecomDigi 2023) and a good profit margin of 17.19%, indicating excellent market penetration and integration.

Maxis is renowned for offering top-notch broadband and mobile services. It is essential to the development of digital enterprise solutions and 5G. Maxis markets itself as a high-end supplier, emphasizing excellent services and a powerful brand identity. It targets wealthy consumers who are prepared to pay for the greatest network experience as well as high-value postpaid users. However, buyers who are price conscious might not find its premium cost appealing. Maxis maintains a strong profit margin of 9.75% despite having the lowest revenue (\$2.2 billion) (Maxis 2024), demonstrating that its emphasis on high-end services and valuable clients is lucrative even with a lesser market share.

Value-added services, pricing, data allotments, network quality, and new products are some of the elements that drive competition in the Malaysian mobile network market. The growth of streaming platforms, online gaming, and digital financial services has increased demand for data-driven services, which is making competition more fierce. To attract and retain customers, operators are concentrating on building tailored services, improving digital content offerings, and extending 5G coverage.

Even while every nation faces unique chances and problems, certain themes come up. The consolidation trend shown by mergers and acquisitions in Malaysia (Celcom and Digi) and Thailand (Dtac and TrueMove H) may result in more rivalry and market concentration. These combined companies' financial results, however, differ. Despite having a bigger market share, TrueMove H in Thailand suffers with profitability, whereas Celcom Digi in Malaysia exhibits excellent integration with significant revenue and profitability. This demonstrates the intricacies of mergers and acquisitions, where post-merger success depends on operational effectiveness, cost control, and brand integration.

TRENDS AND CHALLENGES

1. Trends

- **Technological Leapfrogging:** The ASEAN countries' quick adoption of 4G and 5G technology is an example of a phenomena known as technological leapfrogging, in which developing nations choose more modern, cutting-edge solutions while avoiding older ones. The rising demand for data-intensive apps and the falling cost of mobile devices are the main drivers of this trend.
- **Mobile-First Ecosystem:** High mobile penetration rates, frequently exceeding 100%, indicate a trend toward a mobile-first world in which people primarily communicate and obtain information via mobile devices. Economic growth, social interaction, and digital inclusion are all significantly impacted by this trend.

- **Value-Added Service Differentiation:** With value-added services, operators are differentiating themselves as markets expand and competition intensifies. Their strategy is to increase consumer value and loyalty by combining mobile services with digital content, financial services, and other lifestyle offers.
- **Market Consolidation and Competition:** Intense competition among big operators coexists with market consolidation in the mobile network landscape, as seen by the mergers of TrueMove H and Dtac in Thailand and Celcom and Digi in Malaysia. Consolidation and competition combine to influence consumer choices and market structure.

2. Challenges

- **Bridging the Digital Divide:** Rural and remote areas often lack access to high-speed internet and new mobile technologies, which hinders their participation in the digital economy and worsens existing socioeconomic inequalities.
- **Optimizing Network Quality:** To meet the rising demand for data-heavy applications while maintaining high network quality, continuous investment in infrastructure is essential. This is especially important as 5G networks become more widespread and user expectations for speed and reliability grow.
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- **Sustaining Profitability:** In a highly competitive market with rising operational costs and the need for network upgrades, maintaining profitability is a significant challenge for operators. Balancing competitive pricing, value-added services, and cost optimization is crucial for long-term financial health.
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- **Ensuring Cybersecurity and Data Privacy:** With the increasing reliance on mobile devices and online services, concerns about cybersecurity and data privacy are growing. Operators must proactively address these issues by investing in strong security measures, raising user awareness, and complying with evolving data protection regulations.

CONCLUSION & FUTURE RESEARCH

Conclusion

In conclusion, the four ASEAN countries examined are leading the charge in mobile technology adoption, propelled by rising mobile penetration, swift 4G and 5G network rollouts, and a surging demand for data-driven services. Operators are strategically navigating this dynamic environment by enhancing value-added services, maintaining competitive pricing, and prioritizing network quality to attract and retain customers.

Nonetheless, to achieve sustainable and inclusive growth in the mobile network sector, it is crucial to address the digital divide, ensure consistent network quality, maintain profitability, and safeguard against cybersecurity threats.

Future Research

- **User Behavior and Digital Inclusion:** Further research is needed to understand the nuanced ways in which users in different socioeconomic and demographic groups engage with mobile technologies. This includes examining data usage patterns, preferences for value-added services, and the impact of mobile technology on social inclusion and economic empowerment.
- **5G's Transformative Potential:** Investigating the transformative potential of 5G across various sectors, such as healthcare, education, agriculture, and manufacturing, is crucial for understanding its role in driving economic growth and societal development. This includes assessing the impact of 5G on productivity, innovation, and the creation of new business models.
- **Regulatory Frameworks and Policy Implications:** Analyzing the evolving regulatory frameworks governing the mobile network sector is essential for promoting competition, innovation, and consumer protection. This includes examining spectrum allocation policies, net neutrality regulations, and data privacy laws to ensure a fair and competitive market environment.
- **Sustainability and Environmental Impact:** As the mobile network sector expands, it is important to consider its environmental impact and develop strategies for sustainable development. This includes researching energy-efficient network infrastructure, responsible e-waste management, and the role of mobile technology in promoting environmental sustainability.

By addressing these research areas, policymakers, regulators, and industry stakeholders can gain a deeper understanding of the mobile network landscape in Southeast Asia and develop strategies to promote innovation, competition, and digital inclusion while ensuring environmental sustainability and social responsibility.

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